

Branch and bound

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Solution of the practice quiz

In order to calculate an upper bound, we need a feasible solution. Here, any tour can be selected. For instance, consider the tour $L \rightarrow G \rightarrow B \rightarrow Z \rightarrow L$. Its length is 575. Therefore, we initialize

$$\begin{aligned}x^* &= L \rightarrow G \rightarrow B \rightarrow Z \rightarrow L, \\f^* &= 575.\end{aligned}$$

The tour starts in Lausanne. We create subproblems by branching on the city visited just after Lausanne:

- P_1 : $L \rightarrow B$,
- P_2 : $L \rightarrow G$,
- P_3 : $L \rightarrow Z$.

All these subproblems are feasible. We calculate a lower bound for each of them, using the method described in the problem description. In the following tables, the numbers in bold correspond to an arc already selected. For problem P_1 , the lower bound is 486, based on the following table:

	Arc 1	Arc 2	Sum
L	64	104	168
G	64	158	222
B	104	125	229
Z	125	228	353
			972

For problem P_2 , the bound is also 486.

	Arc 1	Arc 2	Sum
L	64	104	168
G	64	158	222
B	104	125	229
Z	125	228	353
			972

For problem P_3 , the bound is 548.

	Arc 1	Arc 2	Sum
L	64	228	292
G	64	158	222
B	104	125	229
Z	125	228	353
			1096

Note that none of these lower bounds is higher than the best upper bound $f^* = 575$. Therefore, we have to branch. The current set of problems is

$$P, P_1, P_2, P_3.$$

Consider problem P_1 , where it has been decided that Bern will be visited after Lausanne. We create two subproblems, corresponding to the next city:

- P_{11} : $L \rightarrow B \rightarrow Z$.
- P_{12} : $L \rightarrow B \rightarrow G$,

The current set of problems is

$$P, P_1, P_2, P_3, P_{11}, P_{12}.$$

The lower bound for problem P_{11} is 486, based on the following table:

	Arc 1	Arc 2	Sum
L	64	104	168
G	64	158	222
B	104	125	229
Z	125	228	353
			972

This bound is not higher than the best upper bound $f^* = 575$, and the problem cannot be discarded. But the problem is trivial to solve. Indeed, once the first three cities have been chosen, the last one is obvious. Therefore, the solution of P_{11} is $L \rightarrow B \rightarrow Z \rightarrow G \rightarrow L$, with length 572. As it is better than the current upper bound, this is the new candidate for optimality, and we set

$$\begin{aligned} x^* &= L \rightarrow B \rightarrow Z \rightarrow G \rightarrow L, \\ f^* &= 572. \end{aligned}$$

As we have solved problem P_{11} , the current set of problems is

$$P, P_1, P_2, P_3, P_{12}.$$

The lower bound for problem P_{12} is 502.5, based on the following table:

	Arc 1	Arc 2	Sum
L	64	104	168
G	64	158	222
B	104	158	262
Z	125	228	353
			1005

This bound is not higher than the best upper bound $f^* = 572$, and the problem cannot be discarded. But the problem is trivial to solve. Indeed, once the first three cities have been chosen, the last one is obvious. Therefore, the solution of P_{12} is $L \rightarrow B \rightarrow G \rightarrow Z \rightarrow L$, with length 769. As it is higher than the solution of P_{11} , P_{12} can be discarded and the solution of P_1 is the solution of P_{11} . The current set of problems is

$$P, P_2, P_3.$$

Consider now problem P_2 , where it has been decided that Geneva will be visited after Lausanne. We create two subproblems, corresponding to the next city:

- P_{21} : $L \rightarrow G \rightarrow Z$,
- P_{22} : $L \rightarrow G \rightarrow B$.

The current set of problems is

$$P, P_2, P_3, P_{21}, P_{22}.$$

The lower bound on problem P_{21} is 572, based on the following table:

	Arc 1	Arc 2	Sum
L	64	104	168
G	64	279	343
B	104	125	229
Z	125	279	404
			1144

This is equal to the current upper bound f^* . Therefore, there is no point pursuing this problem as it will not lead to a better solution than the one we already have. We discard it, and the set of problem is

$$P, P_2, P_3, P_{22}.$$

The lower bound on P_{22} is 502.5.

	Arc 1	Arc 2	Sum
L	64	104	168
G	64	158	222
B	104	158	262
Z	125	228	353
			1005

This bound is not higher than the best upper bound $f^* = 572$, and the problem cannot be discarded. But the problem is trivial to solve. Indeed, once the first three cities have been chosen, the last one is obvious. Therefore, the solution of P_{22} is $L \rightarrow G \rightarrow B \rightarrow Z \rightarrow L$, with length 575. As P_{21} has been discarded, it is the solution of P_2 . But as it is higher than the current upper bound $f^* = 572$, P_2 is discarded. The current set of problems is

$$P, P_3.$$

Consider now problem P_3 , where it has been decided that Zürich will be visited after Lausanne. We create two subproblems, corresponding to the next city:

- P_{31} : $L \rightarrow Z \rightarrow B$,
- P_{32} : $L \rightarrow Z \rightarrow G$.

The current set of problems is

$$P, P_3, P_{31}, P_{32}.$$

The lower bound of P_{31} is 548, based on the table below.

	Arc 1	Arc 2	Sum
L	64	228	292
G	64	158	222
B	104	125	229
Z	125	228	353
			1096

This bound is not higher than the best upper bound $f^* = 572$, and the problem cannot be discarded. But the problem is trivial to solve. Indeed, once the first three cities have been chosen, the last one is obvious. Therefore, the solution of P_{31} is $L \rightarrow Z \rightarrow B \rightarrow G \rightarrow L$, with length 575. As it is higher than the current upper bound $f^* = 572$, the problem is discarded. The current set of problems is

$$P, P_3, P_{32}.$$

The lower bound on problem P_{32} is 685.5.

	Arc 1	Arc 2	Sum
L	64	228	292
G	64	279	343
B	104	125	229
Z	279	228	507
			1371

As it is higher than the current upper bound $f^* = 572$, the problem is discarded. The solution of P_3 is the solution of P_{31} . P_3 is discarded. And the solution of P is the solution of P_1 :

$$x^* = L \rightarrow B \rightarrow Z \rightarrow G \rightarrow L,$$

$$f^* = 572.$$

The process is illustrated as a tree in Figure 1, where each node corresponds to a subproblem, and its lower bound is indicated. The nodes in gray correspond to the optimal solution.

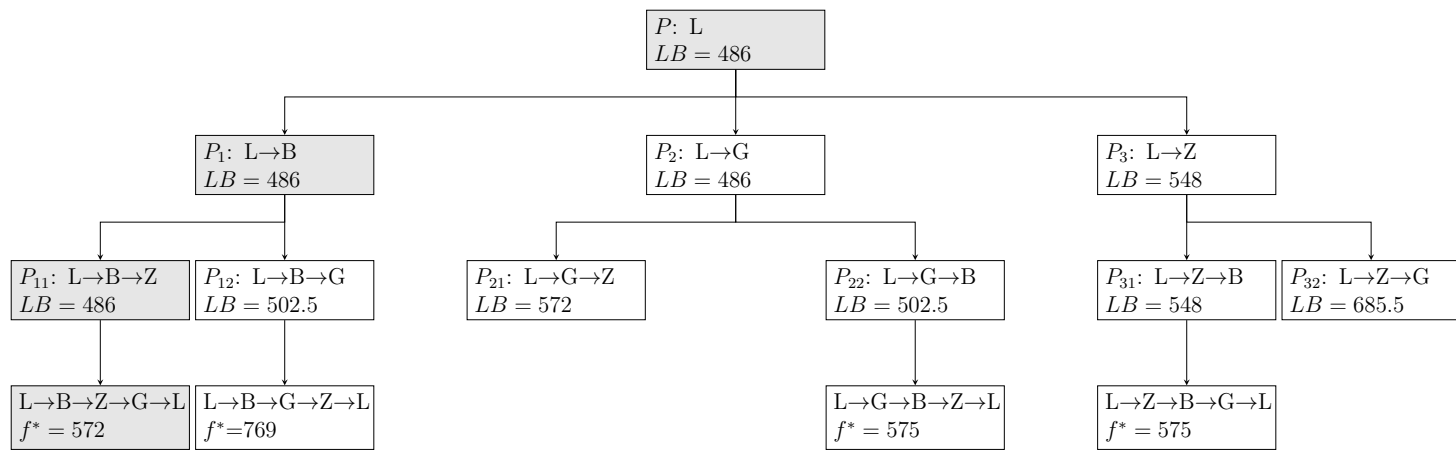


Figure 1: Branch and bound tree